ENME485/691 Industrial Artificial Intelligence
Homework 04 - Fall 2026
Due: October 19 (before 12:30 pm)

Objective: Implement the analytical tools and assess the health of the shaft in a rotor-bearing system.

1 Background
See Homework 02 for background information.

2 Problem Statement
Develop both a **SOM** and **SOM-MQE** method for assessing the health condition of the shaft based on the provided vibration data

3 Data Source
The dataset is the same as the dataset for Homework 03. Use the following files from each folder for your assessment

Training Data:

Healthy – vibration data measured under normal condition (20 observations).

Faulty – (1) vibration data measured under shaft unbalance condition level 1 (1 screw on the disk) (20 observations) **AND** (2) vibration data measured under shaft unbalance condition level 2 (2 screws on the disk) (20 observations).

Testing Data:

Vibration data measured from the testbed with unknown health conditions (30 observations). All health assessment results should come from testing data sets.

4 Suggested Algorithms

Feature Extraction	Extract and add more features from the raw signal (e.g. RMS, peak to peak, standard deviation from the time domain signal, etc.)
Modelling	SOM, SOM-MQE

5 Result Submission
For each **group leader**, please submit a (five to ten minute) presentation file and a report document presenting the health assessment result. The submission report should contain the following sections:

(1) Introduction
   a. background information and description of the problem
   b. raw data description

(2) Implemented Solution

a. brief introduction of the analytics process (sample preparation, feature extraction and selection, application of suggested algorithms)
b. flow chart of entire solution process

(3) Results
a. figures or tables showing the assessment results for both the SOM and SOM-MQE approaches, including the health label of each test sample and a confusion matrix of the test sample results

(4) Discussion
a. include any observations about the modelling techniques, extracted features, results, etc.
b. compare the results with the logistic regression results from Homework 02

(5) Conclusion
a. brief statement concluding the important findings

(6) Appendix – Codes
a. all the programming codes used to achieve the results in the report

6 Appendix - Labels for Testing Data

Use the following to create confusion matrix.

Healthy	Unbalance Level 1	Unbalance Level 2
Data Mar-26-14 Time 1533-21	Mar-26-14 Time 1549-12	Mar-26-14 Time 1606-21
Data Mar-26-14 Time 1533-22	Mar-26-14 Time 1549-13	Mar-26-14 Time 1606-22
Data Mar-26-14 Time 1533-23	Mar-26-14 Time 1549-14	Mar-26-14 Time 1606-23
Data Mar-26-14 Time 1533-24	Mar-26-14 Time 1549-15	Mar-26-14 Time 1606-24
Data Mar-26-14 Time 1533-25	Mar-26-14 Time 1549-16	Mar-26-14 Time 1606-25
Data Mar-26-14 Time 1533-26	Mar-26-14 Time 1549-17	Mar-26-14 Time 1606-26
Data Mar-26-14 Time 1533-27	Mar-26-14 Time 1549-18	Mar-26-14 Time 1606-27
Data Mar-26-14 Time 1533-28	Mar-26-14 Time 1549-19	Mar-26-14 Time 1606-28
Data Mar-26-14 Time 1533-29	Mar-26-14 Time 1549-20	Mar-26-14 Time 1606-29
Data Mar-26-14 Time 1533-30	Mar-26-14 Time 1549-21	Mar-26-14 Time 1606-30